

# AERO2579 - Wind Tunnel Experiment 1

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Wind tunnel experiments are crucial to test different aerofoil sections in a 2-D flow condition. This allows the lifting potential of a 3-D wing to be predicted from the 2-D data. The lift should increase with increasing angle of attack until the stall point, where flow separation starts to occur and a further increase of angle of attack then reduces the lift. The Wright Brothers tested 2D aerofoil sections in the same way you are going to do it.

## REPORTING and SUBMISSION

*This report is to be completed individually and submitted to canvas. Save the document as a PDF using the following naming convention:*

*FAMILYNAME\_SXXXXXXX Wind Tunnel Lab Sem 1 AERO2579.pdf*

*This report is worth 20% of your final grade.*

- *Ensure you change the header to your name and student number (double click on the header).*
- *Ensure that your work is presentable and easy to read, take care of formatting and text/graphic sizes. All graphs must be displayed digitally with the software of your choice.*
- *Marks will be deducted for poor presentation of work.*

## LEARNING OUTCOMES

In this wind tunnel experiment, you will determine the lift characteristics of an aerofoil at constant velocity as a function of angle of attack (i.e. the angle between the chord line and the air stream direction) and compare the measured data to the standard reference values for low-speed flight conditions.

This will be achieved two ways, first you will be using a NACA 0012 aerofoil with pressure taps along the top and bottom surface to generate a pressure distribution. Using the pressure differences between the top and bottom surface, you will determine the coefficient of lift for that angle of attack. The second method will be using the same aerofoil but using the force balance of the wind tunnel to directly get a lift force, which can be converted to a coefficient of lift. The two experimental results you obtain will then be compared with a literature finding to validate your experiments.

It is highly recommended to record your wind tunnel lab and take plenty of photos to either use for evidence in your discussion section or to help remember how the test was conducted.

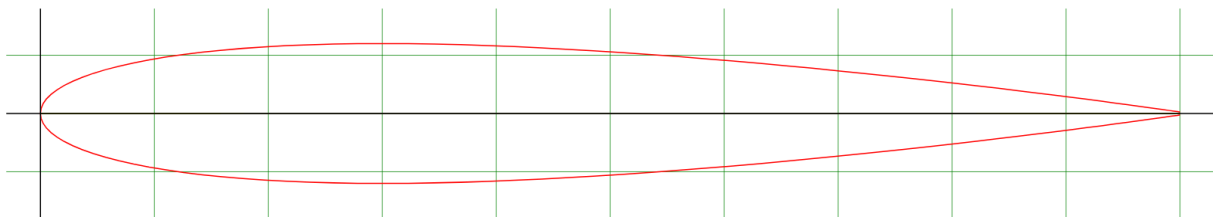


Figure 1: NACA0012 Aerofoil Cross Section

## WIND TUNNEL

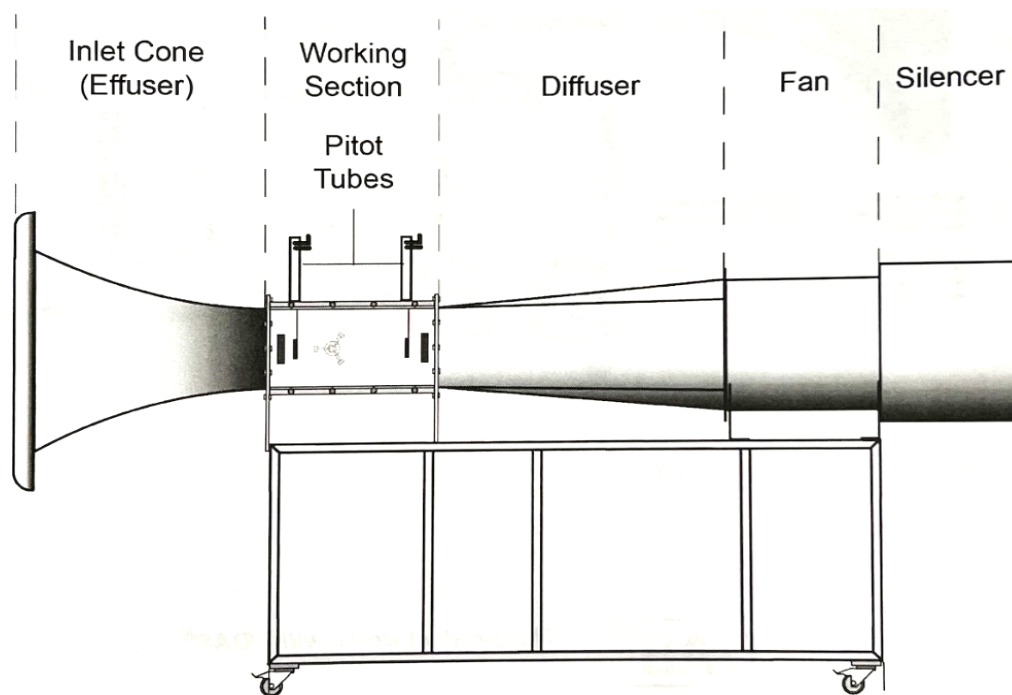


Figure 2: AF1300 General Layout

For this lab you will be using the AF1300 Subsonic Wind Tunnel in building 57 at RMIT (057.02.032). This wind tunnel provides a compact and practical open circuit wind tunnel in which the study of low-speed aerodynamic effects can be analysed. The wind tunnel can be seen in Figure 1 with the different sections identified.

Air enters the tunnel through an aerodynamically designed diffuser (cone) that accelerates the air in a linear manner. It then passes through a honeycomb flow straightener and enters the working section, then passing through a grille before moving through a diffuser and then to the variable speed axial fan. The grille protects the fan from damage by loose objects. The air leaves the fan, passes through a silencer unit and then back out to atmosphere.

The speed of the axial fan (and therefore the air velocity in the working section) is controlled by an electronic drive control in the separate Control and Instrumentation unit. The Control and Instrumentation unit houses facilities for many of the optional ancillaries.

## WORKING SECTION

The working section is of a square section with acrylic roof and floor. The sides are removable full length acrylic panels. The whole unit is supported in an aluminium framework. Each side panel has a special holder to support the optional wind tunnel models. On the top of the working section are holes for the two Pitot devices and two wall tapings to measure the static pressure upstream and downstream of the working section.

## Technical Details

The diagram illustrates the geometry of a hydrofoil cross-section. Key dimensions and features include:

- Upper surface:** The top boundary of the foil.
- Lower surface:** The bottom boundary of the foil.
- Dimensions:**
  - Total length: 137.16 mm
  - Distance from leading edge to the start of the upper surface curve: 38.00 mm
  - Distance from the start of the upper surface curve to the trailing edge: 0.76 mm
  - Distance from the leading edge to the start of the lower surface curve: 41.15 mm
  - Distance from the start of the lower surface curve to the trailing edge: 1.52 mm
  - Distance from the leading edge to the trailing edge: 129.54 mm
  - Distance from the leading edge to the trailing edge (including the gap): 150 mm
- Flow Direction:** Indicated by an arrow pointing left, labeled "Airflow".

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## Acronyms

| Symbol   | Meaning                                                          | Units                                              |
|----------|------------------------------------------------------------------|----------------------------------------------------|
| $h_T$    | Height of water in manometer due to model tapping pressure       | mm <sub>H2O</sub>                                  |
| $\rho_T$ | Model tapping pressure                                           | Pa                                                 |
| $h_w$    | Height of water in manometer due to static wall tapping pressure | mm <sub>H2O</sub>                                  |
| $\rho_w$ | Static wall tapping pressure                                     | Pa                                                 |
| $\rho_a$ | Ambient pressure                                                 | Pa                                                 |
| $C_p$    | Coefficient of pressure                                          | -                                                  |
| $C_L$    | Coefficient of lift                                              | -                                                  |
| $\rho$   | Air density                                                      | kg/m <sup>3</sup> or kg.m <sup>-3</sup>            |
| $L$      | Lift                                                             | N                                                  |
| $S$      | Wing area                                                        | m <sup>2</sup>                                     |
| $R$      | Gas constant                                                     | 287 J kg <sup>-1</sup> K <sup>-1</sup> for dry air |
| $T_a$    | Ambient temperature                                              | K                                                  |
| $v$      | Wind tunnel air velocity                                         | m/s or m.s <sup>-1</sup>                           |
| $c$      | Aerofoil chord                                                   | m                                                  |
| $x$      | A horizontal position (of a set of pressure holes)               | m (or mm where shown)                              |

### Conversions

1 mm H<sub>2</sub>O at 20°C = 9.806 Pa

0°C = 273 K

## Air Density

The density of the air in the wind tunnel can be found from a ratio of the ambient pressure, the gas constant and the ambient temperature:

$$\rho = \frac{P_a}{RT_a} \quad (1)$$

## Two Dimensions

As the aerofoil fully spans the working section of the wind tunnel, the air cannot flow around the wing edges or 'wing tips', as in a real aerofoil. The air flow is only over the upper and lower surface of the wing. This is two-dimensional airflow. Flow around wing tips adds the third dimension, but creates additional drag and other complications, shown by using the shorter span but otherwise identical aerofoil.

## Wing Area S

This is simply the plan view of the aerofoil, which is the product of the chord and the span.

## Derivation of the Coefficient of Lift ( $C_L$ ) Using Force Balance Measurements

It is obvious that a larger wing may give more actual or basic lift than a smaller wing of the same design, so to compare wings of different size: the different scales must be allowed for. Engineers do this by using a non-dimensional value termed the lift coefficient  $C_L$

$$C_L = \frac{L}{\frac{1}{2}\rho v^2 S} \quad (2)$$

The equation shows that the coefficient of lift can be used with the air and aerofoil properties to calculate basic lift.

$$L = C_L \left( \frac{1}{2} \rho v^2 S \right) \quad (3)$$

## Derivation of the Coefficient of Lift ( $C_L$ ) Using Pressure Measurements

### 1. Coefficient of Pressure - $C_p$

Aerofoil lift relies on relative differences in pressure between the atmospheric value around the aircraft and the more local pressures around the aerofoil created as the aerofoil passes through the air (or as the air passes around the aerofoil). As atmospheric pressure changes with weather conditions and aircraft altitude, it is better to allow for these absolute values of pressure when comparing aerofoils. As with the coefficient of lift, engineers have created a non-dimensional form of pressure termed the pressure coefficient  $C_p$ . It is the ratio of the relative pressure difference over a function of the air properties.

$$C_p = \frac{P - P_\infty}{\frac{1}{2} \rho_\infty v_\infty^2} \quad (4)$$

The symbol  $\infty$  means the free stream pressure, density and velocity away from the aerofoil. In the experiment,  $P$  becomes the pressure at the tapping  $P_T$  and  $P_\infty$  becomes the wall static pressure  $P_w$ . The density and velocity are those of the air passing through the working section.

## 2. Finding the Coefficient of Lift

Textbooks show that the coefficient of lift can be found from the area under a chart of the coefficient of pressure against the distance along the aerofoil. The distance along the aerofoil must be made non-dimensional by dividing the position by the chord length, giving a fraction  $x/c$  and

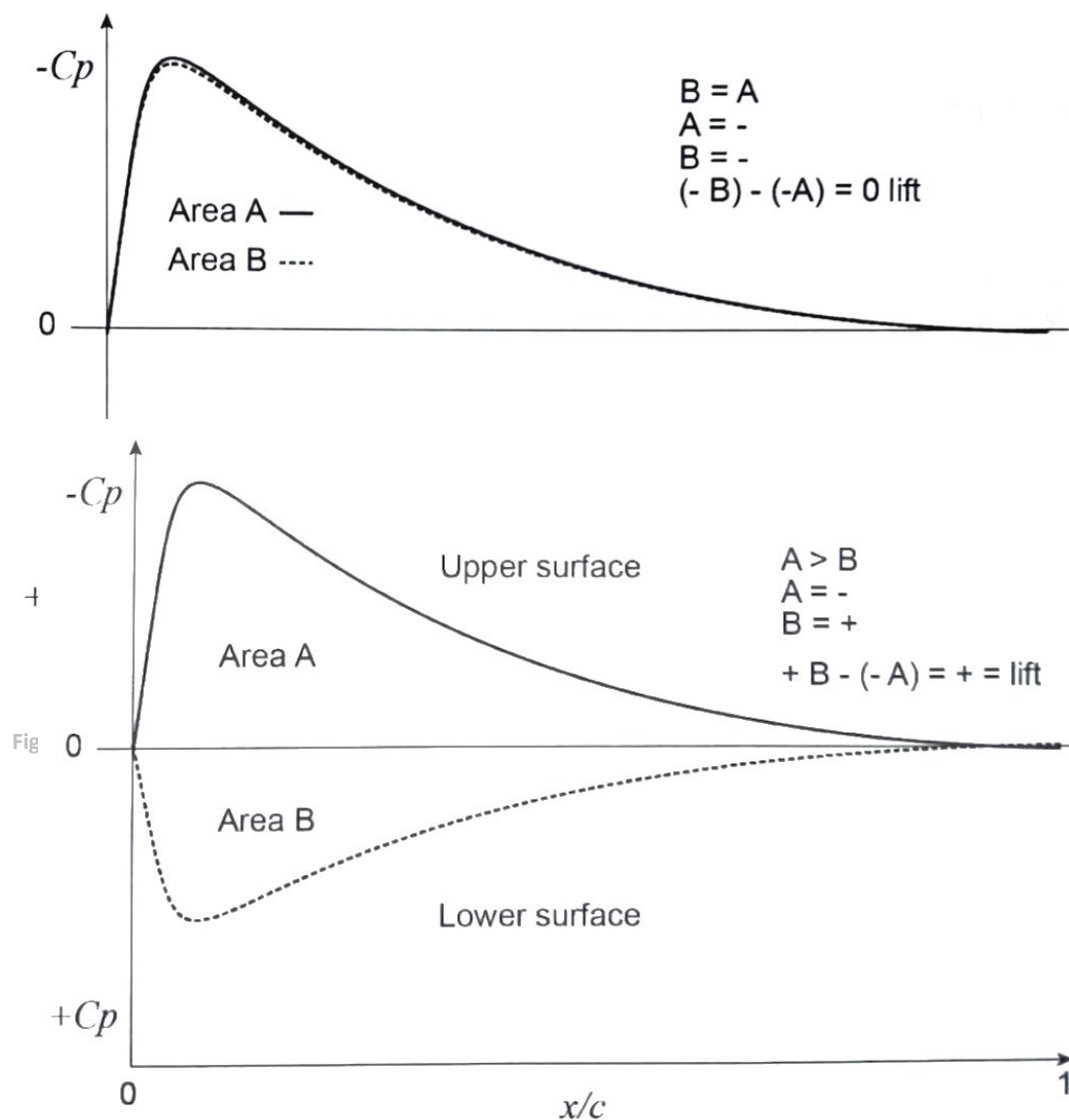


Figure 4: Finding Lift Coefficient

$$C_l = Area\ B - Area\ A \quad (5)$$

Note that the lift coefficient is the difference of the two areas (B-A). For a symmetrical aerofoil at 0 degrees incidence, both areas would be negative and equal, giving a lift of 0 (see Figure 2).

If the area under the upper surface curve is greater and negative, while the lower surface gives positive pressure coefficients, then there is a net positive lift coefficient (see Figure 3).

## Notes

You will be running the experiment and collecting data points to use in the comparison. Use this section to record any notes you have.

## Guidelines for Completing This Task

To achieve the highest possible marks, follow these guidelines:

- Structure answers clearly using bullet points.
- Show all working steps for any required calculations.
- Keep your responses concise—unnecessary details **will not** earn extra marks and may cost you marks if the main explanation is missing as a result of it.
- Use correct technical terms where applicable (e.g., drag, lift, coefficient of lift).
- For the questions asking for a plot or reference from literature make sure its scholarly.
  - [What's a scholarly source?](https://www.youtube.com/watch?v=mmJlV642eTo) (<https://www.youtube.com/watch?v=mmJlV642eTo>)
  - Use google scholar for the search as it will make it easier for you.
  - Try and attain a source that carried out a real-world experimental setup.
  - Hint: We discussed in class some organisation that carried out in-depth airfoil analysis work...

## Discussion Section

1. *Calculate the Reynolds Number and comment on how this would alter your results if you tested at a significantly higher Reynolds Number? (1 mark: answer in 2 dot points)*

- $23^{\circ}\text{C} = 296.15^{\circ}\text{K} \rightarrow 1.828 \times 10^{-5} \text{ Pa} \cdot \text{s}$

$$Re = \frac{\rho_{\infty} V_{\infty} c}{\mu_{\infty}} = \frac{1.20 \times 29.05 \times 0.150}{1.828 \times 10^{-5}} = 286050.328 = 2.861 \times 10^5$$

- Testing a higher Reynolds number would alter the results as there would be an increase in the maximum lift as well as a decrease in the skin friction drag coefficient. With a higher Re, the boundary layer separation is delayed which helps achieve a greater lift coefficient as well as a greater stalling angle of attack. From the equation we can also see for the Re to increase, the inertial forces must be increased relative to viscous forces, which results in a decrease in the skin friction drag.

## Results Section - Lift

1. *Fill in the table with your  $C_l$  results: (Do not add or delete columns. Provide exactly 10 points in your response. Round up to 3 significant figures)*

**Note:**

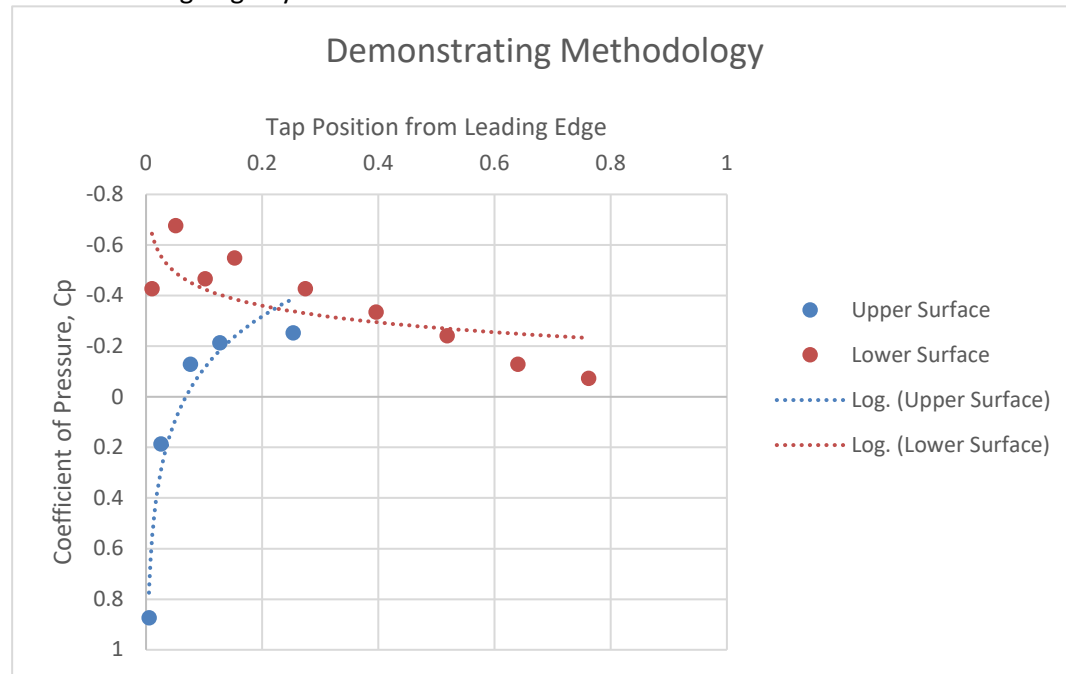
- The coefficient of lift as calculated by using a force balance is automatically calculated and found in the excel file you extracted on the day of the lab experiment.
- Refer to the introductory parts of this document for the calculation of the coefficient of lift using the pressure taps.

| <i>Angle of Attack (AoA) in degrees (0.5 marks)</i>                                                                 | <b>-0.1</b> | <b>2.9</b> | <b>5.9</b> | <b>9.1</b> | <b>11.1</b> | <b>12.1</b> | <b>13.0</b> | <b>14.0</b> | <b>14.9</b> | <b>16.0</b> |
|---------------------------------------------------------------------------------------------------------------------|-------------|------------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <i>Coefficient of Lift Derived using the Force Balance Method (0.5 marks)</i>                                       | -0.012      | 0.010      | 0.299      | 0.573      | 0.709       | 0.673       | 0.766       | 0.806       | 0.772       | 0.718       |
| <i>Coefficient of Lift Derived Using the Pressure Taps Method (4 marks)</i>                                         | -0.106      | 0.170      | 0.405      | 0.671      | 0.773       | 0.837       | 0.882       | 0.910       | 0.933       | 0.910       |
| <i>Percentage Difference Between the Two Lift Coefficients Above (0.5 marks)</i>                                    | 159.32%     | 177.78%    | 30.11%     | 15.76%     | 8.64%       | 21.72%      | 14.08%      | 12.12%      | 18.89%      | 23.59%      |
| <i>Coefficient of lift for the airfoil as reported in the literature under comparable test conditions. (1 mark)</i> | -0.017      | 0.233      | 0.517      | 0.743      | 0.849       | 0.892       | 0.893       | 0.934       | 0.889       | 0.755       |

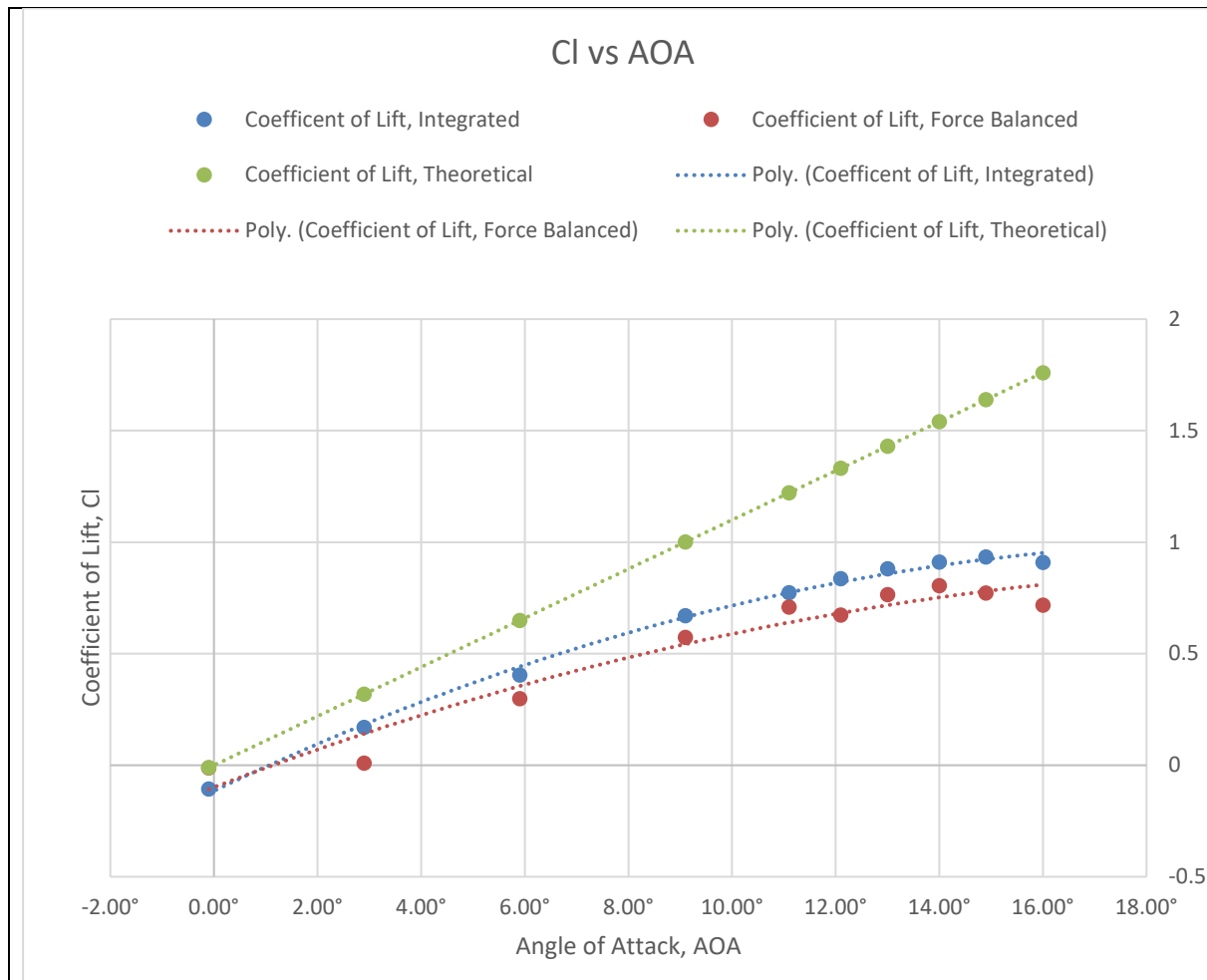
<https://ntrs.nasa.gov/api/citations/19880019495/downloads/19880019495.pdf> page 15

2. For consequential marking, demonstrate the methodology you used to calculate the coefficient of pressure by plotting of the coefficient of pressure at 5.9-degrees AoA graphs you have obtained:

- Using data series 3
- Average each pressure tap 1-20. These are the only ones which are recording the airfoil.
- $C_p = \frac{(P_T \times 1000) - (P_w \times 1000)}{q_\infty}$  (Multiply by 1000 to turn kPa to Pa)
- Map tap locations for both upper and lower surfaces through dividing the distance from the leading edge by the chord of the airfoil



3. Insert a  $c_l$  vs. AoA plot here. Make sure to overlay: [1] the theoretical lift curve slope, [2] the force balance and [3] pressure tap results on the same graph and provide a legend (the coefficient of lift from the force balance in solid lines and the pressure taps in dotted lines): (3.5 marks)



**4. Insert a referenced  $c_l$  vs. AoA plot from literature, making sure to reference your source: (1.5 mark)**

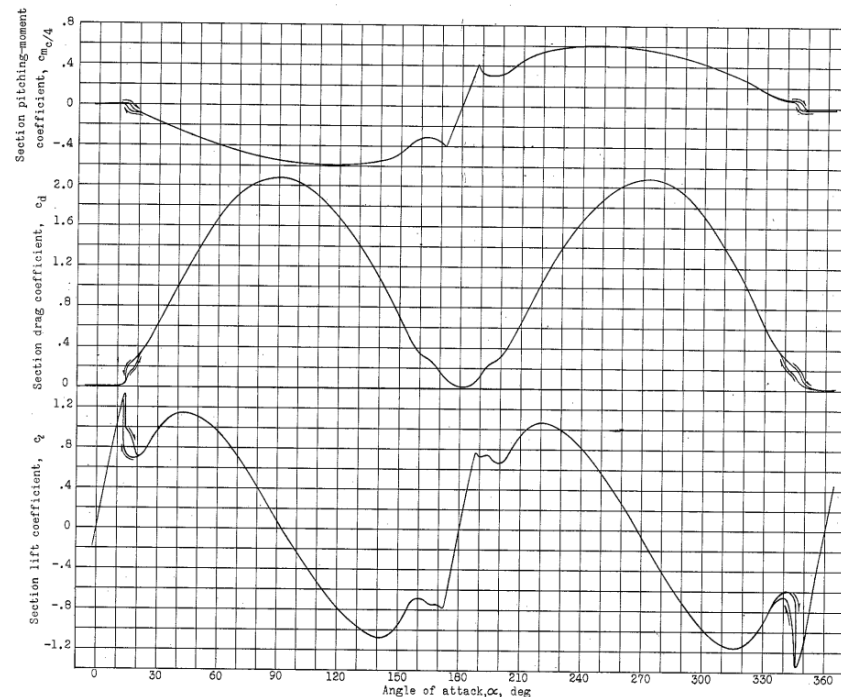


Figure 1.- Aerodynamic characteristics of NACA 0012 airfoil section.  
 $R = 1.8 \times 10^6$ ; airfoil surfaces smooth. Symbols for data points are shown in figure 5.

5. Calculate the lift curve slope of your results. How does this compare with linear aerofoil theory? Explain any differences. (We are looking for three concepts here list them as three dot points) (1.5 marks)

$$C_{L(\text{theoretical})} \rightarrow \alpha_0 = \frac{0.743 - 0.517}{9.1 - 5.9} = 0.071$$

$$C_{L(\text{Force Balance})} \rightarrow \alpha_0 = \frac{0.573 - 0.299}{9.1 - 5.9} = 0.086$$

$$C_{L(\text{Integration})} \rightarrow \alpha_0 = \frac{0.671 - 0.405}{9.1 - 5.9} = 0.083$$

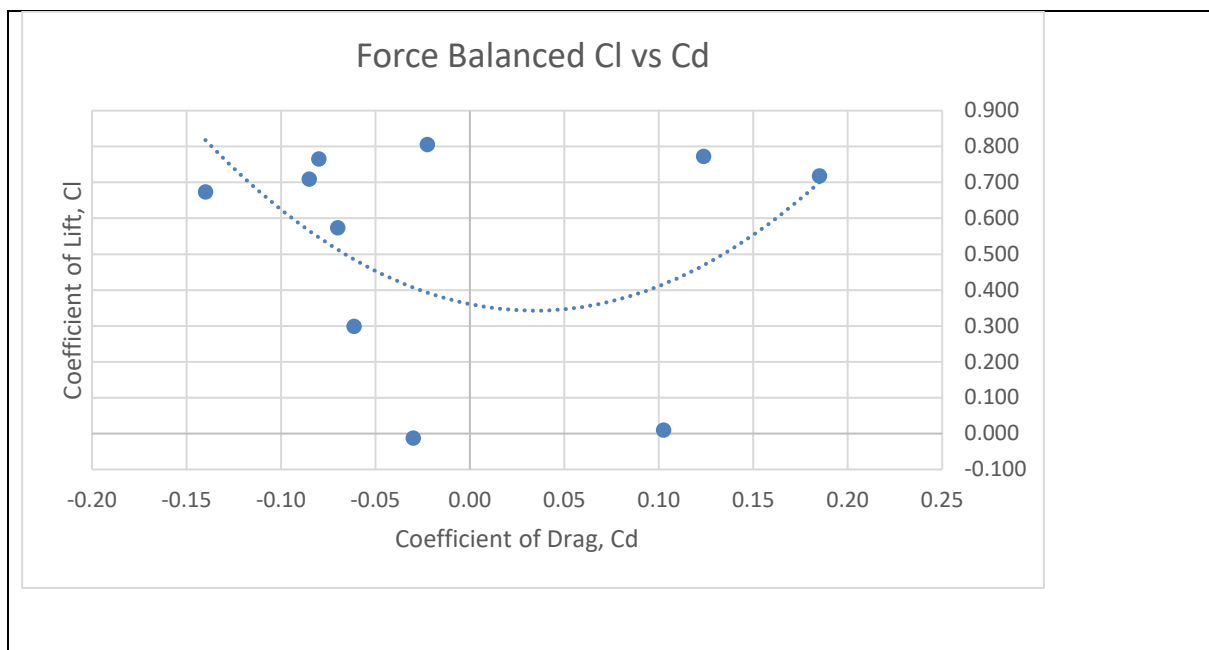
- Linear airfoil theory states that for a thin 2D airfoil in an incompressible flow, the theoretical lift is  $2\pi$  per radian or 0.11 per degree, all our lifts found from this experiment are less than the perfect value, which we would expect in a wind tunnel.
- Viscous v Inviscid flow. With the linear airfoil theory, we presume inviscid flow, no friction, however in reality such as with the experiment there is friction as well as other forces, causing a boundary layer to form, which leads to a lower than theoretically perfect slope.
- Reynolds number. The calculated Reynolds number of  $2.861 \times 10^5$  is on the lower end of  $Re$ 's calculated for this NACA airfoil when looking at NASA's data. NASA's data also shows us that a greater Reynolds number translates to a later separation, and a greater coefficient of lift. In the linear airfoil theory, the perfect slope is greater than what our Reynolds number produces at 0.11 per degree.
- The wind tunnel dimensions are also an interference with measurements that additionally add it variations in the data recorded compared to what is expected in the linear airfoil theory. Where the airfoil meets the walls creates points where the flow fails to simulate a two-dimensional airfoil, creating potential points of leakage where the measured lift may be impacted, which effects the lift slope.

**6. Looking at the percentage differences between the force balance and the pressure taps, why are there differences and where is the greatest difference? (1 mark: answer in 2 dot points)**

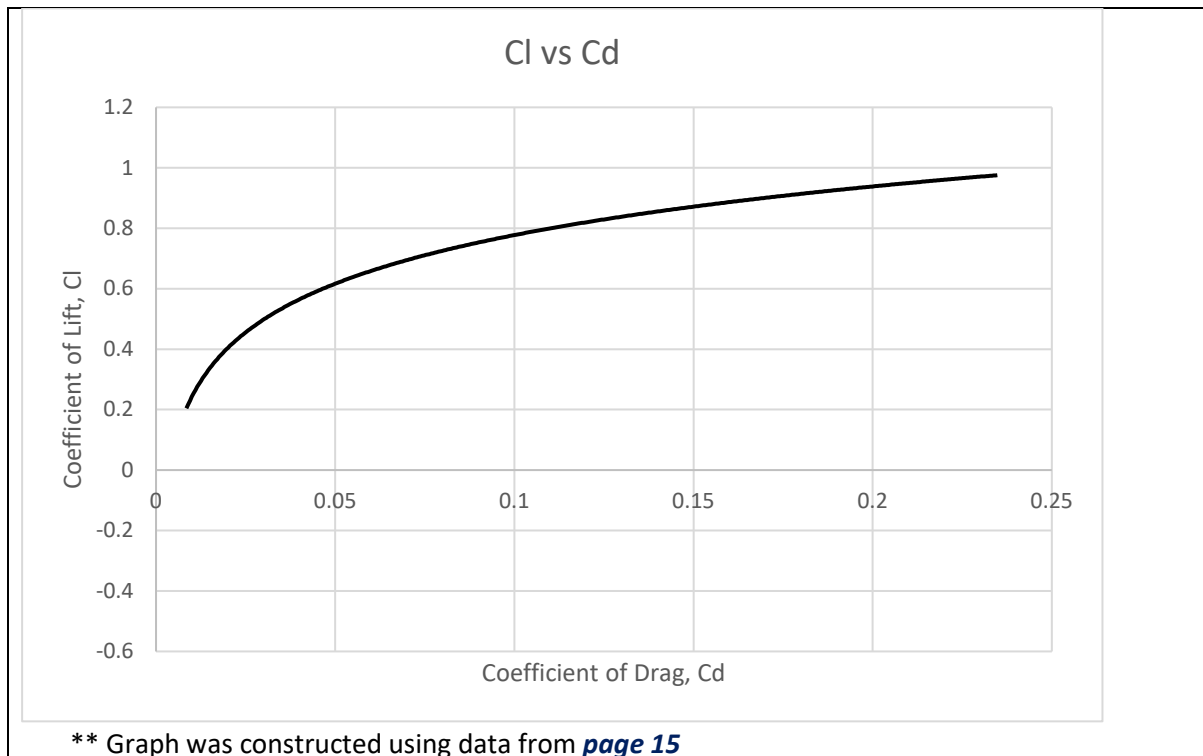
- When calculating the coefficient of lift using the force balance method, the total resultant aerodynamic forces acting on the airfoil are measured. This may also include pressure lift and friction from the airfoil, as well as other external variables which aren't accounted for in the data. However, using the pressure taps, only the lift generated from the normal surface pressure is measured. This may also have some errors occur during integration as well as at certain points of the airfoil.
- The greatest percentage difference is an angle of attack of  $2.9^\circ$  with a difference of 177.78%. Looking at the differences the greatest differences occur when the angle of attack is very low or negative.

## Results Section - Drag

**1. Insert a  $c_l$  vs.  $c_d$  plot using the force balance data in your labs excel spreadsheet: (1.5 marks)**



**2. Insert a referenced  $c_l$  vs.  $c_d$  plot from literature: (1 mark)**



**3. Describe how your data compares with the results from literature? Ensure you mention both Lift and Drag. What factors could account for any differences? (2.5 marks: answer in 5 dot points)**

- The experimental data obtained from force balancing records negative data for most of the tested angle of attacks with the smallest being -0.14. This is vastly different from the coefficient of drag obtained from NASA's measurements which only records positive values, the smallest being 0.00847.
- The Reynolds Number for the two graphs differ slightly as the experimental one was calculated to be  $2.861 \times 10^5$ , the Re from NASA's plot is greater at  $0.70 \times 10^6$ . With our lower Re, we can expect an earlier flow separation as well as a smaller maximum lift coefficient. This can be seen as NASA's maximum coefficient of lift was recorded as 0.9391 at a  $14.10^\circ$  AOA, whereas ours was 0.8056 at a  $14.00^\circ$  AOA.
- With our lab, fluctuations occur which don't fit the trend of the given series. This can be seen in the second data series where there is a spike of +0.07 above the series average. These outliers can most likely be a sign of external noise such as mechanical movements or fluctuations with sensors.
- Both the lab and NASA's tests showed a consistent increase in the coefficient of lift at smaller AOAs, showing relatively linear growth. However, before fluctuations start the experiments peak was 0.8056, whereas NASA reached 0.9391.
- After reaching each airfoil's respective coefficient of lift, both were pitched at a similar AOA, with NASA being at  $14.10^\circ$  and the experiments at  $14.00^\circ$ . Where this occurs, both airfoils record a spike in their coefficient of drag, suggesting that both have a very similar stall point.

## Annotations

Ladson, C L. 1988, *Effects of Independent Variation of Mach and Reynolds Numbers on the Low-Speed Aerodynamic Characteristics of the NACA 0012 Airfoil Section*, CSCL 01B H1/01, NASA, viewed <<https://ntrs.nasa.gov/api/citations/19880019495/downloads/19880019495.pdf>>.

Critzos, C C., Heyson, H H. & Boswinkle, R W., Jr *AERODYNAMIC CHARACTERISTICS OF NACA 0012 AIRFOIL SECTION AT ANGLES OF ATTACK FROM 00 TO 180 °*, viewed <<https://ntrs.nasa.gov/api/citations/19930084501/downloads/19930084501.pdf>>.